

Project Initialization and Planning Phase

Date	29 September 2026
Team ID	SWTID-2026-1393
Project Title	EduGenie – AI Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal aims to support students with an AI-powered learning assistant that makes study material easier to understand, revise and practice. EduGenie uses generative AI to provide explanations, summaries, question answering, quizzes and personalized learning support.

Project Overview	
Objective	The primary objective is to provide students with an accessible AI learning assistant that can explain topics clearly, generate useful study content and support faster, more personalized learning.
Scope	The project covers AI-assisted explanations, summaries/notes, question answering, quiz generation and learning support through a web-based interface.
Problem Statement	
Description	Students often spend significant time searching for suitable explanations and practice material. Existing resources may be scattered, lengthy or not personalized to the learner's immediate question or level.
Impact	Addressing these issues can reduce study time, make difficult concepts easier to understand, improve access to practice material and support a more organized learning experience.
Proposed Solution	
Approach	Employing generative AI through the EduGenie web application to analyze a student's topic or question and generate clear explanations, summaries, quizzes and personalized learning assistance.
Key Features	• AI-generated explanations for difficult topics. • Summary and study-note generation. • Question answering and clarification. • Quiz generation for practice. • Personalized learning support and revision guidance.

Proposed Solution – Resource Requirements

Key Features – continued

- Interactive learning assistance through a web interface.
- AI-generated content based on the student's input.
- Practice-oriented quiz generation.
- Modular design so additional learning features can be added later.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Computer/laptop capable of running the Python application and browser; internet connection for AI API access.
	Memory	Minimum 8 GB RAM recommended for development.
	Storage	Sufficient local storage for source code, dependencies and project documentation.
Software	Programming / Framework	Python, FastAPI
	Frontend	HTML5, CSS3, JavaScript
	Libraries / Tools	Project Python modules and required dependencies; Git for version control.
	Development Environment	Visual Studio Code / suitable Python development environment.
AI / API	Generative AI service	Google Gemini API / Gemini model for explanations, summaries, quizzes and learning assistance.
Data	Application input	Student-provided topics, questions and learning requests; generated responses are used to support the learning workflow.

Team Members – Group 1

No.	Member Name	Group No.
1	BALAMURUGAN M	1
2	Aakash A	1
3	Akash S	1
4	Santhosh kumar R	1
5	Gunaseelan K	1